greenhouse gases, but they are also devastatingly effective in destroying the protective ozone layer.

You look more closely at the center of the South American continent, which—as you know by now—is a vast rain forest. Every night you see thousands of fires. In the daytime, you find the region covered with smoke. Over the years, all over the planet, you find less and less forest and more and more scrub desert.

You look down on the large island of Madagascar. The rivers are colored brown, generating a vast stain in the surrounding ocean. This is topsoil being washed out to sea at a rate so high that in another few decades there will be none left. The same thing is happening, you note, at the mouths of rivers all over the planet.

But no topsoil means no agriculture. In another century, what will they eat? What will they breathe? How will they cope with a changing and more dangerous environment?

From your orbital perspective, you can see that something has unmistakably gone wrong. The dominant organisms, whoever they are—who have gone to so much trouble to rework the surface—are simultaneously destroying their ozone layer and their forests, eroding their topsoil, and performing massive, uncontrolled experiments on their planet's climate. Haven't they noticed what's happening? Are they oblivious to their fate? Are they unable to work together on behalf of the environment that sustains them all?

Perhaps, you think, it's time to reassess the conjecture that there's intelligent life on Earth.

LOOKING FOR LIFE ELSEWHERE: A CALIBRATION

Spacecraft from the Earth have now flown by dozens of planets, moons, comets, and asteroids—equipped with cameras, instruments for measuring heat and radio waves, spectrometers to determine composition, and a host of other devices. We have found not a hint of life anywhere else in the Solar System. But you might be skeptical about our ability to detect life elsewhere, especially life different from the kind we know. Until